AUDIT REPORT

# last-mile-dispatch: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

| Item | Value |
|---|---|
| Repository | https://github.com/kavya2693/last-mile-dispatch |
| Numbers taken from commit | `c7de8a6ad78ef3961f773e55ae9af197085c3f9c` (outputs last changed in `c7de8a6 (2026-10-13 17:55:02 +0400)`) |
| Video | `docs/demo/demo.mp4` · 87.0 s · 1920x1080 · 30 fps · silent |
| Generated | 2026-10-14 18:37 |
| Automated checks | 68 pass · 0 fail · 1 manual (of 69 figures) |
| Assumptions | 13 (A-01 to A-13) |

## How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

### Regenerate

```
make run                                  # data/ (seed 7) and outputs/*.csv, summary.json, run_log.txt (~3 min)
python3 docs/demo/source/tools/export_series.py docs/demo/source/assets/run_series.json
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_rider_board.py docs/demo/source/assets/rider_board.json
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/rider_board.json docs/demo/source/compositions/value.html --name SHEET
for f in s02-hook s04-routes s07-end; do python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/run_series.json docs/demo/source/compositions/$f.html --name SERIES; done
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q looks -o renders/last-mile-dispatch-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/last-mile-dispatch-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks, MD/DOCX/PDF
```

# 1. Figures

| ID | Shown | Slides | Source | Column / row / key | How computed | Type | Check |
|---|---|---|---|---|---|---|---|
| F-01 | 3 industry chips | 01 | author's framing | - | - | general knowledge | manual: Context statement, not a repo output. |
| F-02 | 4 problem cards | 01 | README 'The problem', 'Insights and learning' | README text | - | general knowledge | ✓ verified |
| F-03 | 26 hubs | 02, 05, 07 | outputs/summary.json; data/hubs.csv | hubs; row count | - | direct | ✓ verified |
| F-04 | 473 riders | 02, 05 | data/riders.csv | row count | - | direct | ✓ verified |
| F-05 | 477,868 orders | 02, 05 | data/orders.csv.gz | row count | - | direct | ✓ verified |
| F-06 | 6 weeks / 42 days | 02, 05 | data/days.csv | row count = 42 | 42 / 7 = 6 weeks | derived | ✓ verified |
| F-07 | 30 minutes to each door / 30-minute promise | 02, 07 | src/last_mile_dispatch/config.py:27 | SLA_MIN = 30 | - | direct | ✓ verified |
| F-08 | riders sent out every 2 min / EVERY 2 MINUTES / dispatch wave 2 min | 02, 03, 05 | src/last_mile_dispatch/config.py:23 | WAVE_MIN = 2 | - | direct | ✓ verified |
| F-09 | Ramadan · weekends · rain | 02 | data/days.csv | ramadan, weekend, rain columns | any ramadan, any weekend, exactly one rain day | direct | ✓ verified |
| F-10 | 36 riders sometimes skip doors | 02 | data/riders.csv | remote_marker == True | count | direct | ✓ verified |
| F-11 | hub dots + circles | 02, 10 | data/hubs.csv; config.py:12 (SERVICE_RADIUS_KM = 4.0) | lat/lon -> km from the centroid (demo assets/run_series.json) | - | direct | ✓ verified |
| F-12 | ORDERS → BAG → MATCH → DOOR CHECK → A/B TEST → VERDICT | 03 | src/last_mile_dispatch/ | functions | - | direct | ✓ verified |
| F-13 | 4 WEEKS / over 4 weeks / 4 weeks | 03, 08, 09 | outputs/summary.json | experiment_days = 28 | 28 / 7 = 4 | derived | ✓ verified |
| F-14 | cheaper orders / fewer cancels / fewer fake marks | 03 | outputs/ab_readout.csv | welch_hi for cost_per_delivered, cancel_rate, fake_accepted_per_1k | welch_hi < 0 for all three | derived | ✓ verified |
| F-15 | 3 steps / ≤ 3 stops · 1 km | 04, 05 | src/last_mile_dispatch/bagging.py:3-5, :51-87; config.py:37, :65-67 | BAG_CAPACITY = 3, BAG_MAX_DROP_GAP_KM = 1.0, BAG_MAX_CREATED_GAP_MIN = 12.0, BAG_MAX_EXTRA_MIN = 9.0 | - | direct | ✓ verified |
| F-16 | 9.71 | 04 | outputs/ablation.csv | policy=baseline, cost_per_delivered | - | direct | ✓ verified |
| F-17 | 8.95 AED | 04 | outputs/ablation.csv | policy=bagging, cost_per_delivered | - | direct | ✓ verified |
| F-18 | generate_data.py, runner.py run_all(), simulate.py simulate_hub_day(), bagging.py, assignment.py (min_cost, fifo), geo.py, geofence.py, experiment.py, report.py | 05 | src/last_mile_dispatch/, scripts/ | defs | - | direct | ✓ verified |
| F-19 | power.csv, ab_readout.csv, geofence_eval.csv, ablation.csv, rerandomisation.csv, hub_day_metrics.csv | 05 | outputs/ | files | - | direct | ✓ verified |
| F-20 | 3,640 rows | 05 | outputs/hub_day_metrics.csv | row count | 26 x 28 x 5 | direct | ✓ verified |
| F-21 | 11,566 marks | 05 | outputs/geofence_eval.csv | slice=all, events | - | direct | ✓ verified |
| F-22 | hub-day runs 4,004 | 05 | outputs/run_log.txt line 2 | 'simulated 4,004 hub-day-policy runs' | 26 x 14 + 26 x 28 x 5 = 4,004 | direct | ✓ verified |
| F-23 | full run, laptop 180 s | 05 | outputs/summary.json | runtime_s = 179.8 | rounded to whole seconds | derived | ✓ verified |
| F-24 | cancel after 45 min | 05 | src/last_mile_dispatch/config.py:28 | CANCEL_AFTER_MIN = 45 | - | direct | ✓ verified |
| F-25 | 150 m geofence | 05 | src/last_mile_dispatch/config.py:79 | GEOFENCE_RADIUS_M = 150.0 | - | direct | ✓ verified |
| F-26 | 500 re-draws / Test re-run 500 times | 05, 09 | outputs/rerandomisation.csv | draws | - | direct | ✓ verified |
| F-27 | 13 vs 13 hubs | 05, 08, 09 | outputs/summary.json | arms.treatment, arms.control | len of each list | direct | ✓ verified |
| F-28 | 4 parallel workers · 5 | 05 | src/ | --workers default 4; | - | direct | ✓ verified |

| ID | Shown | Slides | Source | Column / row / key | How computed | Type | Check |
|---|---|---|---|---|---|---|---|
| | policies, 1 seed | | last_mile_dispatch/cli.py:29; runner.py:36-41; config.py:62 | same seed for every policy; ALL_POLICIES | | | |
| F-29 | CI slice 8 hubs × 7 d | 05 | src/last_mile_dispatch/cli.py:38, :40; .github/workflows/ci.yml | hubs.head(8), experiment days head(7) | - | direct | ✓ verified |
| F-30 | generate_data.py · seed 7 | 05 | scripts/generate_data.py:168 | --seed default 7 | - | direct | ✓ verified |
| F-31 | two lines, Feb 2 - Mar 1 | 06 | outputs/hub_day_metrics.csv | sum(cancelled) / sum(orders) by date and policy | injected SERIES block must equal the recomputation | derived | ✓ verified |
| F-32 | RAIN DAY · 42.5% → 14.0% | 06 | outputs/stress_split.csv | day_type=rain day, cancel_rate | x 100 | direct | ✓ verified |
| F-33 | 37,530 → 11,063 | 06 | outputs/hub_day_metrics.csv | sum(cancelled) by policy | - | direct | ✓ verified |
| F-34 | −70.5% | 06 | outputs/hub_day_metrics.csv | F-33 | (11,063 / 37,530 - 1) x 100 = -70.52% | derived | ✓ verified |
| F-35 | 125,715 orders · 9.71 → 9.36 · −3.7% | 06 | outputs/stress_split.csv | day_type=ordinary weekday | cost_treat / cost_base - 1 | derived | ✓ verified |
| F-36 | 184,449 orders · 9.65 → 8.30 · −14.0% | 06 | outputs/stress_split.csv | day_type=weekend or Ramadan | cost_treat / cost_base - 1 | derived | ✓ verified |
| F-37 | 16,817 orders · 10.42 → 6.84 · −34.4% | 06 | outputs/stress_split.csv | day_type=rain day | cost_treat / cost_base - 1 | derived | ✓ verified |
| F-38 | RAMADAN band from Feb 18 | 06 | data/days.csv | first date with ramadan = True | - | direct | ✓ verified |
| F-39 | Rider 01-05 rows | 07 | demo assets/rider_board.json (re-run of the repo's simulator), injected into compositions/value.html | rows | km = sum of road legs; waited = wave time - oldest order's creation; due = earliest creation + 30 min | direct | ✓ verified |
| F-40 | 795 of 821 delivered, 7 cancelled · old way: 59 cancelled (old way = FIFO, one order per trip) | 07 | outputs/hub_day_metrics.csv | hub_id=18, date=2026-02-14, policy in {bagging+matching+geofence, baseline} | - | direct | ✓ verified |
| F-41 | + 3 bags held for the next wave · 5 free riders, 8 bags waiting | 07 | demo assets/rider_board.json | bags_held, free_riders, bags_waiting_before | held = bags waiting - riders matched | direct | ✓ verified |
| F-42 | All 9 inside the 30-minute promise. | 07 | demo assets/rider_board.json | orders_dispatched; stops[].on_time | every stop ETA - creation <= 30 min | direct | ✓ verified |
| F-43 | HUB 18 | 07 | outputs/summary.json | arms.treatment | - | direct | ✓ verified |
| F-44 | −10.6% | 08 | outputs/ab_readout.csv | metric=cost_per_delivered, rel_diff | x 100 | direct | ✓ verified |
| F-45 | AED 9.79 → 8.75 | 08 | outputs/ab_readout.csv | metric=cost_per_delivered, control, treatment | - | direct | ✓ verified |
| F-46 | +28.3 | 08 | outputs/ab_readout.csv | metric=on_time_share_of_orders, diff | x 100 | direct | ✓ verified |
| F-47 | 58.0% → 86.2% | 08 | outputs/ab_readout.csv | metric=on_time_share_of_orders, control, treatment | x 100 | direct | ✓ verified |
| F-48 | 3.7% · control 11.6% | 08 | outputs/ab_readout.csv | metric=cancel_rate, treatment, control | x 100 | direct | ✓ verified |
| F-49 | −72.4% | 08 | outputs/ab_readout.csv | metric=fake_accepted_per_1k, rel_diff | x 100 | direct | ✓ verified |
| F-50 | 11.7 → 3.2 per 1,000 | 08 | outputs/ab_readout.csv | metric=fake_accepted_per_1k, control, treatment | - | direct | ✓ verified |
| F-51 | 326,981 orders | 08 | outputs/summary.json | orders_in_experiment | - | direct | ✓ verified |
| F-52 | on all four, the 95% confidence range of the difference stays clear of zero (Welch t-test) | 08 | outputs/ab_readout.csv | welch_lo, welch_hi | lo > 0 or hi < 0 for cost, on-time, cancel, fake | derived | ✓ verified |
| F-53 | 0.667 / 0.650 | 09 | outputs/ab_readout.csv | metric=utilization, control, treatment | - | direct | ✓ verified |
| F-54 | gap strip from -0.040 to +0.007, crossing zero | 09 | outputs/ab_readout.csv | metric=utilization, welch_lo, welch_hi | - | direct | ✓ verified |
| F-55 | 95% range, p = 0.16 | 09 | outputs/ab_readout.csv | metric=utilization, welch_p | - | direct | ✓ verified |
| F-56 | 27 · TOO FEW / utilisation needs 27, the test had 13 | 09 | outputs/power.csv | metric=utilization, hubs_per_arm_needed | - | direct | ✓ verified |
| F-57 | track -0.05..+0.05 = 260 px; range 25 -> 148 px (left 25, width 123), estimate dot centred at 87 px (left 75), zero at 130 px | 09 | compositions/s06-twist.html; outputs/ab_readout.csv | welch_lo, welch_hi, diff | px = (v + 0.05) x 2,600 | derived | ✓ verified |

| ID | Shown | Slides | Source | Column / row / key | How computed | Type | Check |
|---|---|---|---|---|---|---|---|
| F-62 | -0.017 | 09 | outputs/ab_readout.csv | metric=utilization, diff | - | direct | ✓ verified |
| F-63 | The simulator's true drop, all 26 hubs: -0.026 | 09 | outputs/ab_readout.csv | metric=utilization, true_effect_all_hubs | - | direct | ✓ verified |
| F-64 | Cancellations 2 · Cost per order 5 · ENOUGH | 09 | outputs/power.csv | metric in {cancel_rate, cost_per_delivered}, hubs_per_arm_needed | ENOUGH = needed <= 13 | direct | ✓ verified |
| F-65 | to catch the effect 4 times in 5 | 09 | src/last_mile_dispatch/experiment.py:77, :85 | power: float = 0.8 | - | direct | ✓ verified |
| F-66 | 97-99% of the time · false alarms 2-7% | 09 | outputs/rerandomisation.csv | welch_coverage, aa_false_positive_welch | min and max over metrics, x 100, rounded | derived | ✓ verified |
| F-67 | THE FIX, IN THE REPO · Size the test before trusting it · The run report prints this table next to the A/B results | 09 | src/last_mile_dispatch/experiment.py:77-90, :127; cli.py:78-98; tests/test_experiment.py:47, :53, :66; outputs/run_report.md | functions, tests, report sections | - | direct | ✓ verified |
| F-68 | Dashed line = 13, what this test had | 09 | outputs/summary.json | len(arms.treatment) | - | direct | ✓ verified |
| F-69 | NEXT · Let the roster shrink with the new dispatch: the fewest riders that still keep orders on time. | 09 | README 'Insights and learning' (last paragraph) and 'Known limitations' (the biggest gap: the roster is fixed) | README text | - | direct | ✓ verified |
| F-58 | MIT | 10 | LICENSE line 1 | - | - | direct | ✓ verified |
| F-59 | ~3 min | 10 | outputs/summary.json | runtime_s = 179.8 | 179.8 / 60 = 3.0 min | derived | ✓ verified |
| F-60 | SciPy · pandas · NumPy · Python | 10 | pyproject.toml dependencies | numpy, pandas, scipy | - | direct | ✓ verified |
| F-61 | $ make run  # 26 hubs, 4 weeks, ~3 min | 10 | Makefile; outputs/summary.json | run target; hubs; experiment_days | - | direct | ✓ verified |

## 2. Automated check run (2026-10-14 18:37)

| ID | Expected on slide | Recomputed from repo | Result |
|---|---|---|---|
| F-01 | 3 industry chips | - | manual |
| F-02 | true | true | ✓ pass |
| F-03 | 26 | 26 | ✓ pass |
| F-04 | 473 | 473 | ✓ pass |
| F-05 | 477,868 | 477,868 | ✓ pass |
| F-06 | 42 days | 42 days | ✓ pass |
| F-07 | true | true | ✓ pass |
| F-08 | true | true | ✓ pass |
| F-09 | true | true | ✓ pass |
| F-10 | 36 | 36 | ✓ pass |
| F-11 | true | true | ✓ pass |
| F-12 | true | true | ✓ pass |
| F-13 | 4 weeks | 4 weeks | ✓ pass |
| F-14 | true | true | ✓ pass |
| F-15 | true | true | ✓ pass |
| F-16 | 9.71 | 9.71 | ✓ pass |
| F-17 | 8.95 | 8.95 | ✓ pass |
| F-18 | true | true | ✓ pass |
| F-19 | true | true | ✓ pass |
| F-20 | 3,640 | 3,640 | ✓ pass |
| F-21 | 11,566 | 11,566 | ✓ pass |
| F-22 | true | true | ✓ pass |
| F-23 | 180 s | 180 s | ✓ pass |
| F-24 | true | true | ✓ pass |
| F-25 | true | true | ✓ pass |
| F-26 | 500 | 500 | ✓ pass |
| F-27 | 13 vs 13 | 13 vs 13 | ✓ pass |
| F-28 | true | true | ✓ pass |
| F-29 | true | true | ✓ pass |
| F-30 | true | true | ✓ pass |
| F-31 | true | true | ✓ pass |
| F-32 | 42.5% → 14.0% | 42.5% → 14.0% | ✓ pass |
| F-33 | 37,530 → 11,063 | 37,530 → 11,063 | ✓ pass |
| F-34 | -70.5% | -70.5% | ✓ pass |
| F-35 | 125,715 · 9.71 → 9.36 · -3.7% | 125,715 · 9.71 → 9.36 · -3.7% | ✓ pass |
| F-36 | 184,449 · 9.65 → 8.30 · -14.0% | 184,449 · 9.65 → 8.30 · -14.0% | ✓ pass |
| F-37 | 16,817 · 10.42 → 6.84 · -34.4% | 16,817 · 10.42 → 6.84 · -34.4% | ✓ pass |
| F-38 | 2026-02-18 | 2026-02-18 | ✓ pass |
| F-39 | true | true | ✓ pass |
| F-40 | 795 of 821 delivered, 7 cancelled · FIFO 59 | 795 of 821 delivered, 7 cancelled · FIFO 59 | ✓ pass |
| F-41 | 3 held · 5 riders, 8 bags | 3 held · 5 riders, 8 bags | ✓ pass |

| ID | Expected on slide | Recomputed from repo | Result |
|---|---|---|---|
| F-42 | true | true | ✓ pass |
| F-43 | true | true | ✓ pass |
| F-44 | -10.6% | -10.6% | ✓ pass |
| F-45 | 9.79 → 8.75 | 9.79 → 8.75 | ✓ pass |
| F-46 | +28.3 | +28.3 | ✓ pass |
| F-47 | 58.0% → 86.2% | 58.0% → 86.2% | ✓ pass |
| F-48 | 3.7% · control 11.6% | 3.7% · control 11.6% | ✓ pass |
| F-49 | -72.4% | -72.4% | ✓ pass |
| F-50 | 11.7 → 3.2 | 11.7 → 3.2 | ✓ pass |
| F-51 | 326,981 | 326,981 | ✓ pass |
| F-52 | true | true | ✓ pass |
| F-53 | 0.667 / 0.650 | 0.667 / 0.650 | ✓ pass |
| F-54 | -0.040 to +0.007 | -0.040 to +0.007 | ✓ pass |
| F-55 | 0.16 | 0.16 | ✓ pass |
| F-56 | 27 | 27 | ✓ pass |
| F-57 | true | true | ✓ pass |
| F-62 | -0.017 | -0.017 | ✓ pass |
| F-63 | -0.026 | -0.026 | ✓ pass |
| F-64 | 2 / 5 | 2 / 5 | ✓ pass |
| F-65 | true | true | ✓ pass |
| F-66 | 97-99% · 2-7% | 97-99% · 2-7% | ✓ pass |
| F-67 | true | true | ✓ pass |
| F-68 | 13 | 13 | ✓ pass |
| F-69 | true | true | ✓ pass |
| F-58 | true | true | ✓ pass |
| F-59 | ~3 min | ~3 min | ✓ pass |
| F-60 | true | true | ✓ pass |
| F-61 | true | true | ✓ pass |

## 3. Assumptions register

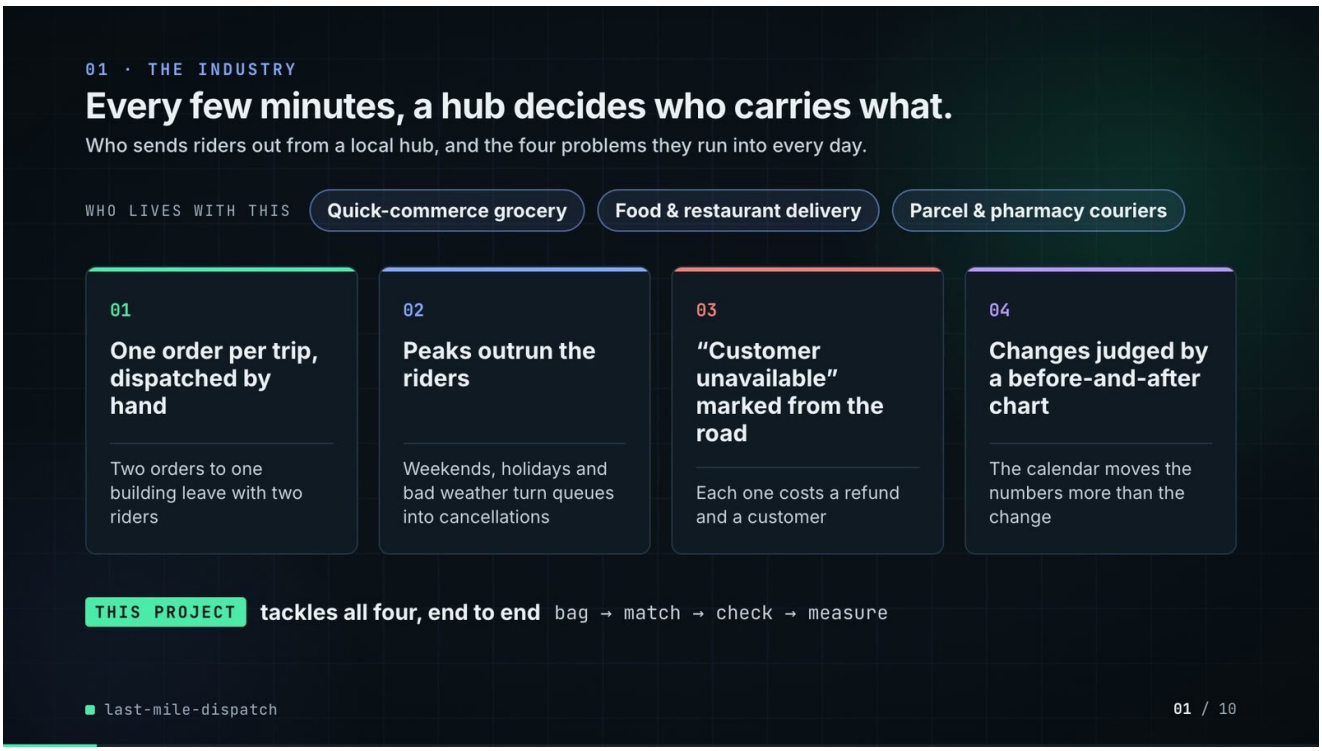
| ID | Assumption | Source | Figures affected | Slides affected |
|---|---|---|---|---|
| A-01 | All data is synthetic, produced by a seeded generator sized to the production problem (26 hubs, 473 riders, 42 days, 477,868 orders). Production dispatch data is proprietary. | scripts/generate_data.py:9, :168 (seed 7); src/last_mile_dispatch/config.py:11-18; README 'The data' | all figures | slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10 |
| A-02 | The rider roster is fixed and sized for an ordinary weekday evening; every policy pays for the same riders. Cost savings therefore come mostly from cancelled orders turned into deliveries, and utilisation falls. | scripts/generate_data.py:84-85 (make_riders); README 'Known limitations' (the biggest gap) | F-04, F-33, F-34, F-35, F-36, F-37, F-44, F-45, F-53, F-54, F-55, F-56, F-62, F-63, F-69 | slides 02, 05, 06, 08, 09 |
| A-03 | Travel is straight-line distance x 1.35 at an hour-dependent speed (bike 24 km/h, car 22 km/h; peaks x0.68-0.88, rain x0.6), not a road graph; riders return to the hub between trips. | src/last_mile_dispatch/config.py:13, :36; geo.py:29-46; simulate.py:247-249; README 'Known limitations' | F-32, F-33, F-34, F-35, F-36, F-37, F-39, F-44, F-45, F-46, F-47 | slides 06, 07, 08 |
| A-04 | Matching prices each rider-bag pair in AED: km cost of the vehicle, 1.0 per minute late, minus 1.4 per minute already waited, minus a 9.0 credit per order. The matcher is myopic (one 2-minute wave at a time). | src/last_mile_dispatch/config.py:74-76; assignment.py:46-58; README 'Known limitations' | F-14, F-33, F-34, F-39, F-44, F-45, F-46, F-47, F-48 | slides 03, 06, 07, 08 |
| A-05 | Bagging limits: drops within 1 km and created within 12 minutes of the seed order, at most 3 stops, and the first customer may wait at most 9 extra minutes. | src/last_mile_dispatch/config.py:37, :65-67; bagging.py:51-87 | F-15, F-16, F-17, F-39, F-41 | slides 04, 05, 07 |
| A-06 | Geofence rule: a 'customer unavailable' mark is flagged when fewer than 2 of the rider's last 6 GPS pings are within 150 m of the customer; flagged riders are sent back to the door (treatment) or the flag is only logged (baseline). | src/last_mile_dispatch/config.py:43, :79-80; geofence.py:31-39; simulate.py:218-230 | F-21, F-25, F-49, F-50 | slides 05, 08 |
| A-07 | Fake marks: about 8% of riders (36 of 473 in the data) mark 'customer unavailable' without reaching the door on 22% of their stops; fakes are known by construction and riders do not change habit when flagged. | src/last_mile_dispatch/config.py:40-41; scripts/generate_data.py:94; README 'Known limitations' | F-10, F-21, F-49, F-50 | slides 02, 05, 08 |
| A-08 | Service promise 30 minutes from order creation; a customer cancels if nothing has left the hub after 45 minutes; the dispatcher wakes every 2 minutes; 2 minutes handover per stop. | src/last_mile_dispatch/config.py:23, :27-29 | F-07, F-08, F-24, F-32, F-33, F-34, F-42, F-46, F-47, F-48 | slides 02, 03, 05, 06, 07, 08 |
| A-09 | Experiment design: hubs sorted by pre-period orders per day, paired, one coin flip per pair (seed 2026); one number per hub over four weeks; Welch t and hub-bootstrap 95% intervals; power from the pre-period between-hub spread. | src/last_mile_dispatch/cli.py:22, :58-98; experiment.py:20-124 | F-27, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-52, F-53, F-54, F-55, F-56, F-57, F-62, F-63, F-64, F-65, F-66, F-67, F-68 | slides 05, 08, 09 |
| A-10 | Calendar: two pre-period weeks from 2026-01-19, then four experiment weeks 2026-02-02 to 2026-03-01; weekends are Saturday-Sunday; Ramadan from 2026-02-18; one rain day 2026-02-07 (35% more | src/last_mile_dispatch/config.py:16-18; scripts/generate_data.py:35-36; README 'The data' | F-06, F-09, F-13, F-31, F-32, F-35, F-36, F-37, F-38, F-40 | slides 02, 03, 05, 06, 07, 08, 09 |

| ID | Assumption | Source | Figures affected | Slides affected |
|---|---|---|---|---|
| | orders, 40% slower traffic). | | | |
| A-11 | The rider board (slide 07) re-runs the repo's own simulator for hub 18 on 2026-02-14 under the treatment and records the 21:46 wave; drop ETAs replay the simulator's clock with geo.travel_min for riders who are honest and customers who are home. | `docs/demo/source/tools/export_rider_board.py; simulate.py:189-253` | F-39, F-40, F-41, F-42 | slides 07 |
| A-12 | Video rounding: percentages to 1 decimal, AED to 2 decimals, utilisation and CI bounds to 3 decimals, counts with thousands separators; '~3 min' and '180 s' are rounded run times. | `this audit (display convention)` | F-16, F-17, F-23, F-32, F-34, F-35, F-36, F-37, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-53, F-54, F-55, F-59, F-62, F-63, F-66 | slides 04, 05, 06, 08, 09, 10 |
| A-13 | Slide 01 (industries, everyday problems) is general delivery-operations knowledge used as context. It is not a result of this repo and carries no numbers. | `author's framing; README 'The problem' supports problems 1-4` | F-01, F-02 | slides 01 |

## 4. Slide by slide

### Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`




#### On-screen text (verbatim)

01 · THE INDUSTRY

Every few minutes, a hub decides who carries what.

Who sends riders out from a local hub, and the four problems they run into every day.

WHO LIVES WITH THIS · Quick-commerce grocery · Food & restaurant delivery · Parcel & pharmacy couriers

01 One order per trip, dispatched by hand — Two orders to one building leave with two riders

02 Peaks outrun the riders — Weekends, holidays and bad weather turn queues into cancellations

03 "Customer unavailable" marked from the road — Each one costs a refund and a customer

04 Changes judged by a before-and-after chart — The calendar moves the numbers more than the change

THIS PROJECT tackles all four, end to end · bag → match → check → measure

#### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-01 | 3 industry chips | `author's framing · -` | - | manual |
| F-02 | 4 problem cards | `README 'The problem', 'Insights and learning'` · README text | - | ✓ |

Assumptions: A-13

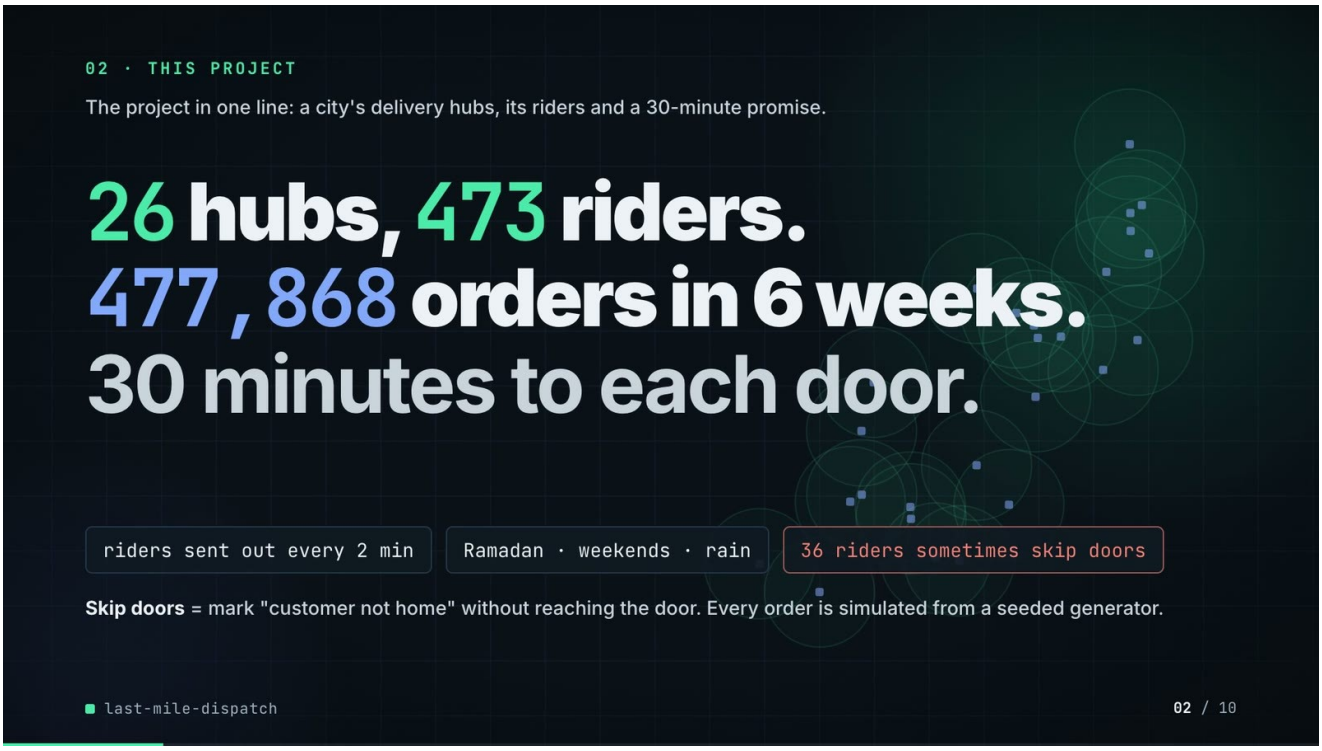
**Illustrative, not data:**

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

## Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html (SERIES block injected by tools/inject_sheet.py)`




### On-screen text (verbatim)

02 · THIS PROJECT

The project in one line: a city's delivery hubs, its riders and a 30-minute promise.

26 hubs, 473 riders. 477,868 orders in 6 weeks. 30 minutes to each door.

riders sent out every 2 min · Ramadan · weekends · rain · 36 riders sometimes skip doors

Skip doors = mark "customer not home" without reaching the door. Every order is simulated from a seeded generator.

### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-03 | 26 hubs | `outputs/summary.json; data/hubs.csv` · hubs; row count | - | ✓ |
| F-04 | 473 riders | `data/riders.csv` · row count | - | ✓ |
| F-05 | 477,868 orders | `data/orders.csv.gz` · row count | - | ✓ |
| F-06 | 6 weeks / 42 days | `data/days.csv` · row count = 42 | 42 / 7 = 6 weeks | ✓ |
| F-07 | 30 minutes to each door / 30-minute promise | `src/last_mile_dispatch/config.py:27` · SLA_MIN = 30 | - | ✓ |
| F-08 | riders sent out every 2 min / EVERY 2 MINUTES / dispatch wave 2 min | `src/last_mile_dispatch/config.py:23` · WAVE_MIN = 2 | - | ✓ |
| F-09 | Ramadan · weekends · rain | `data/days.csv` · ramadan, weekend, rain columns | any ramadan, any weekend, exactly one rain day | ✓ |
| F-10 | 36 riders sometimes skip doors | `data/riders.csv` · remote_marker == True | count | ✓ |
| F-11 | hub dots + circles | `data/hubs.csv; config.py:12 (SERVICE_RADIUS_KM = 4.0)` · lat/lon -> km from the centroid (demo assets/run_series.json) | - | ✓ |

Assumptions: A-01, A-07, A-08, A-10

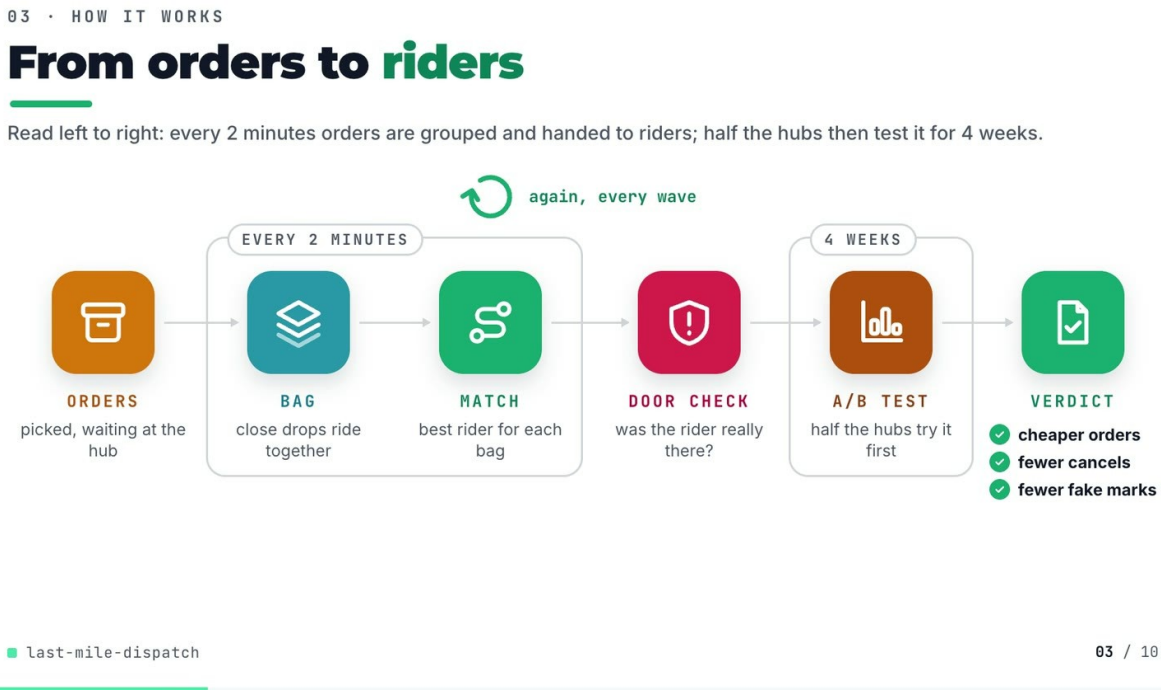
**Illustrative, not data:**

- The background is the 26 real hub positions with a 4 km service circle each (decoration, low opacity).

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; counts come from data/*.csv.

## Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`




### On-screen text (verbatim)

03 · HOW IT WORKS

From orders to riders

Read left to right: every 2 minutes orders are grouped and handed to riders; half the hubs then test it for 4 weeks.

ORDERS picked, waiting at the hub → [EVERY 2 MINUTES: BAG close drops ride together → MATCH best rider for each bag] → DOOR CHECK was the rider really there? → [4 WEEKS: A/B TEST half the hubs try it first] → VERDICT ✓ cheaper orders ✓ fewer cancels ✓ fewer fake marks

again, every wave (loop over MATCH)

### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-08 | riders sent out every 2 min / EVERY 2 MINUTES / dispatch wave 2 min | `src/last_mile_dispatch/config.py:23` · WAVE_MIN = 2 | - | ✓ |
| F-12 | ORDERS → BAG → MATCH → DOOR CHECK → A/B TEST → VERDICT | `src/last_mile_dispatch/` · functions | - | ✓ |
| F-13 | 4 WEEKS / over 4 weeks / 4 weeks | `outputs/summary.json` · experiment_days = 28 | 28 / 7 = 4 | ✓ |
| F-14 | cheaper orders / fewer cancels / fewer fake marks | `outputs/ab_readout.csv` · welch_hi for cost_per_delivered, cancel_rate, fake_accepted_per_1k | welch_hi < 0 for all three | ✓ |

Assumptions: A-05, A-06, A-08, A-09

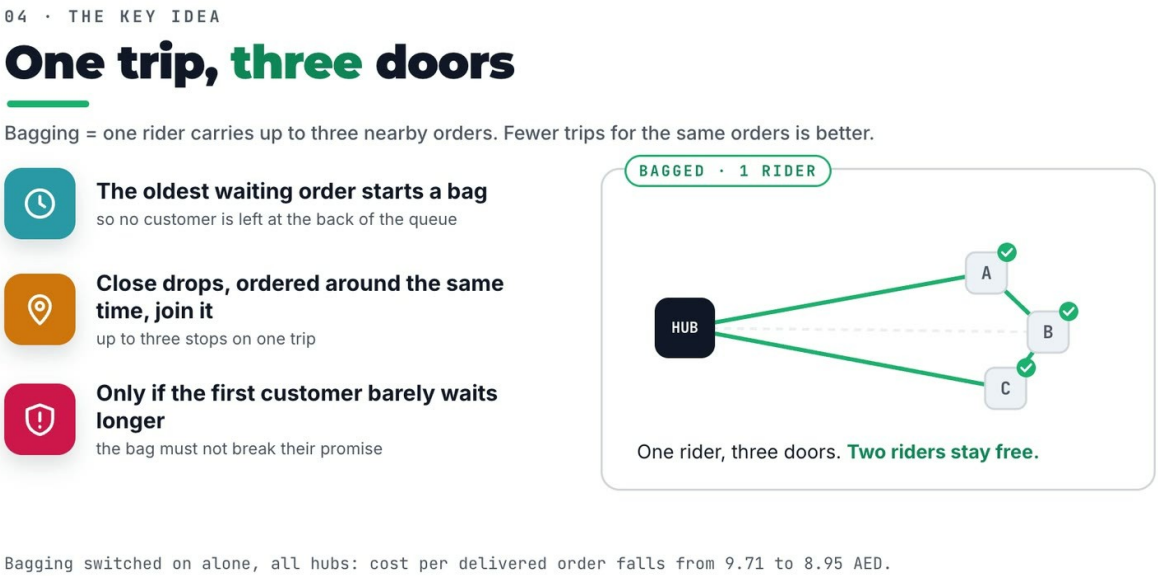
**Illustrative, not data:**

- Icons and colours are a visual language, not data.

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth comes from ab_readout.csv (F-14).

## Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition `compositions/explainer.html (#ex-b ...)`




### On-screen text (verbatim)

04 · THE KEY IDEA
One trip, three doors
Bagging = one rider carries up to three nearby orders. Fewer trips for the same orders is better.
The oldest waiting order starts a bag — so no customer is left at the back of the queue
Close drops, ordered around the same time, join it — up to three stops on one trip
Only if the first customer barely waits longer — the bag must not break their promise
ONE ORDER PER TRIP · 3 RIDERS → BAGGED · 1 RIDER (HUB, A, B, C). One rider, three doors. Two riders stay free.
Bagging switched on alone, all hubs: cost per delivered order falls from 9.71 to 8.95 AED.

### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-15 | 3 steps / ≤ 3 stops · 1 km | `src/last_mile_dispatch/bagging.py:3-5, :51-87; config.py:37, :65-67` · BAG_CAPACITY = 3, BAG_MAX_DROP_GAP_KM = 1.0, BAG_MAX_CREATED_GAP_MIN = 12.0, BAG_MAX_EXTRA_MIN = 9.0 | - | ✓ |
| F-16 | 9.71 | `outputs/ablation.csv` · policy=baseline, cost_per_delivered | - | ✓ |
| F-17 | 8.95 AED | `outputs/ablation.csv` · policy=bagging, cost_per_delivered | - | ✓ |

Assumptions: A-02, A-05, A-12

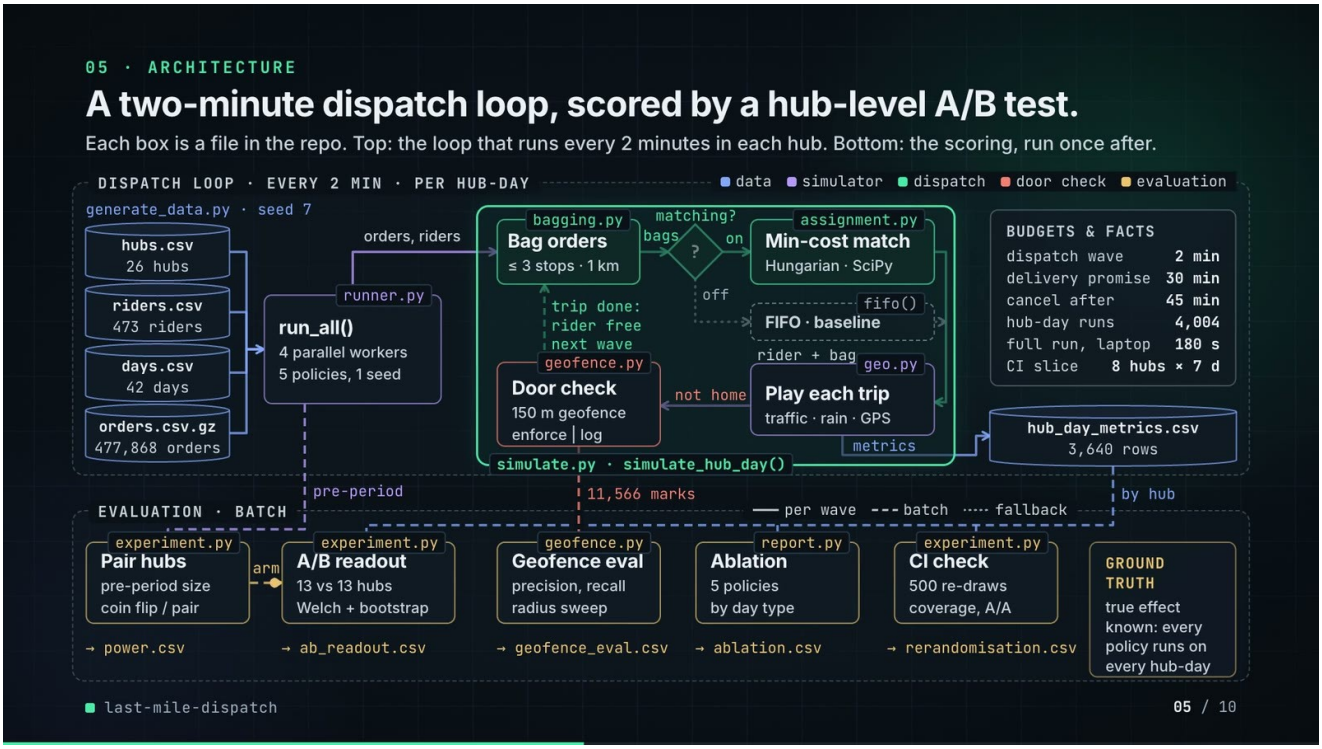
**Illustrative, not data:**

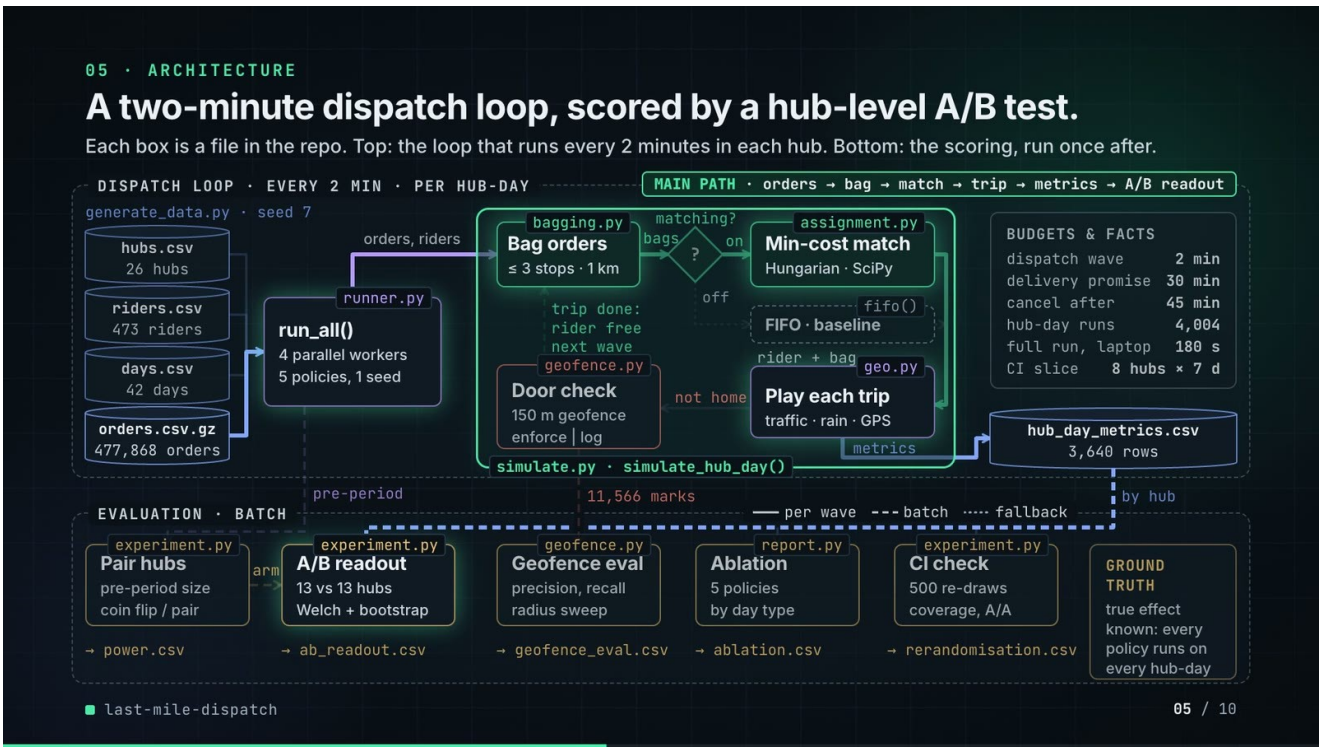
- The HUB/A/B/C demo is schematic: it shows the rule in bagging.make_bags, not a real bag.

To change this slide: Markup under #ex-b in compositions/explainer.html; the footnote figures are F-16 and F-17.

## Slide 05: Architecture

26.0-41.0 s · composition `compositions/s03-architecture.html`




### On-screen text (verbatim)

05 · ARCHITECTURE

A two-minute dispatch loop, scored by a hub-level A/B test.

Each box is a file in the repo. Top: the loop that runs every 2 minutes in each hub. Bottom: the scoring, run once after.

Legend (on the lane borders): data · simulator · dispatch · door check · evaluation; per wave (solid) · batch (dashed) · fallback (dotted)

DISPATCH LOOP · EVERY 2 MIN · PER HUB-DAY — generate_data.py · seed 7: hubs.csv 26 hubs, riders.csv 473 riders, days.csv 42 days, orders.csv.gz 477,868 orders

runner.py run_all(): 4 parallel workers, 5 policies, 1 seed → orders, riders →

simulate.py · simulate_hub_day(): bagging.py Bag orders (≤ 3 stops · 1 km) → bags → matching? → on → assignment.py Min-cost match (Hungarian · SciPy); off → fifo() FIFO · baseline → rider + bag → geo.py Play each trip (traffic · rain · GPS) → not home → geofence.py Door check (150 m geofence · enforce | log) → trip done: rider free, next wave

metrics → hub_day_metrics.csv 3,640 rows · BUDGETS & FACTS: dispatch wave 2 min, delivery promise 30 min, cancel if not out 45 min, hub-day runs 4,004, full run, laptop 180 s, CI slice 8 hubs × 7 d (cancel row reads 'cancel after 45 min')

EVALUATION · BATCH: experiment.py Pair hubs (pre-period size, coin flip / pair) → arm → experiment.py A/B readout (13 vs 13 hubs, Welch + bootstrap) → ab_readout.csv; geofence.py Geofence eval (precision, recall, radius sweep) ← 11,566 marks → geofence_eval.csv; report.py Ablation (5 policies, by day type) → ablation.csv; experiment.py CI check (500 re-draws, coverage, A/A) → rerandomisation.csv; → power.csv; GROUND TRUTH: true effect known: every policy runs on every hub-day

MAIN PATH · orders → bag → match → trip → metrics → A/B readout

**Figures on this slide**

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-03 | 26 hubs | `outputs/summary.json; data/hubs.csv` · hubs; row count | - | ✓ |
| F-04 | 473 riders | `data/riders.csv` · row count | - | ✓ |
| F-05 | 477,868 orders | `data/orders.csv.gz` · row count | - | ✓ |
| F-06 | 6 weeks / 42 days | `data/days.csv` · row count = 42 | 42 / 7 = 6 weeks | ✓ |
| F-08 | riders sent out every 2 min / EVERY 2 MINUTES / dispatch wave 2 min | `src/last_mile_dispatch/config.py:23` · WAVE_MIN = 2 | - | ✓ |
| F-15 | 3 steps / ≤ 3 stops · 1 km | `src/last_mile_dispatch/bagging.py:3-5, :51-87; config.py:37, :65-67` · BAG_CAPACITY = 3, BAG_MAX_DROP_GAP_KM = 1.0, BAG_MAX_CREATED_GAP_MIN = 12.0, BAG_MAX_EXTRA_MIN = 9.0 | - | ✓ |
| F-18 | generate_data.py, runner.py run_all(), simulate.py simulate_hub_day(), bagging.py, assignment.py (min_cost, fifo), geo.py, geofence.py, experiment.py, report.py | `src/last_mile_dispatch/, scripts/` · defs | - | ✓ |
| F-19 | power.csv, ab_readout.csv, geofence_eval.csv, ablation.csv, rerandomisation.csv, hub_day_metrics.csv | `outputs/` · files | - | ✓ |
| F-20 | 3,640 rows | `outputs/hub_day_metrics.csv` · row count | 26 x 28 x 5 | ✓ |
| F-21 | 11,566 marks | `outputs/geofence_eval.csv` · slice=all, events | - | ✓ |
| F-22 | hub-day runs 4,004 | `outputs/run_log.txt line 2` · 'simulated 4,004 hub-day-policy runs' | 26 x 14 + 26 x 28 x 5 = 4,004 | ✓ |
| F-23 | full run, laptop 180 s | `outputs/summary.json` · runtime_s = 179.8 | rounded to whole seconds | ✓ |
| F-24 | cancel after 45 min | `src/last_mile_dispatch/config.py:28` · CANCEL_AFTER_MIN = 45 | - | ✓ |
| F-25 | 150 m geofence | `src/last_mile_dispatch/config.py:79` · GEOFENCE_RADIUS_M = 150.0 | - | ✓ |
| F-26 | 500 re-draws / Test re-run 500 times | `outputs/rerandomisation.csv` · draws | - | ✓ |
| F-27 | 13 vs 13 hubs | `outputs/summary.json` · arms.treatment, arms.control | len of each list | ✓ |
| F-28 | 4 parallel workers · 5 policies, 1 seed | `src/last_mile_dispatch/cli.py:29; runner.py:36-41; config.py:62` · --workers default 4; same seed for every policy; ALL_POLICIES | - | ✓ |
| F-29 | CI slice 8 hubs × 7 d | `src/last_mile_dispatch/cli.py:38, :40; .github/workflows/ci.yml` · hubs.head(8), experiment days head(7) | - | ✓ |
| F-30 | generate_data.py · seed 7 | `scripts/generate_data.py:168` · --seed default 7 | - | ✓ |

Assumptions: A-01, A-03, A-04, A-05, A-06, A-08, A-09

**Illustrative, not data:**

- Packets moving along the wires are decoration in execution order (the dispatch loop turns twice).
- 'per wave' (solid) vs 'batch' (dashed) marks what runs inside each 2-minute wave and what runs after all hub-days.
- The '?' diamond is the decision point Policy.matching (config.py); the dotted path is the FIFO fallback used by the baseline.
- The main-path highlight is the chain that produces the headline A/B numbers; other boxes dim to 75% opacity, which keeps their text above 4.5:1.

To change this slide: Node boxes (#n-*), cylinders (CYL) and the W wires object in compositions/s03-architecture.html.

## Slide 06: The system running

40.6-53.2 s · composition `compositions/s04-routes.html (SERIES block injected by tools/inject_sheet.py from tools/export_series.py)`







### On-screen text (verbatim)

06 · THE SYSTEM RUNNING
Four weeks, 26 hubs, same orders, two dispatchers.
Share of each day's orders that were cancelled, all 26 hubs. Lower is better.
today: one order per trip vs bag + match + door check · RAMADAN band · RAIN DAY · 42.5% → 14.0%
Orders cancelled in 4 weeks: one order per trip 37,530 · bag + match + check 11,063 · −70.5%
(phase B) Fleet cost per delivered order in AED, by type of day. Lower is better: grey = today, green = new dispatch.
WHERE THE SAVING COMES FROM · COST PER DELIVERED ORDER, AED: Ordinary weekday (125,715 orders) 9.71 → 9.36 −3.7% · Weekend or Ramadan (184,449 orders) 9.65 → 8.30 −14.0% · Rain day (16,817 orders) 10.42 → 6.84 −34.4%

### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-31 | two lines, Feb 2 - Mar 1 | outputs/hub_day_metrics.csv · sum(cancelled) / sum(orders) by date and policy | injected SERIES block must equal the recomputation | ✓ |
| F-32 | RAIN DAY · 42.5% → 14.0% | outputs/stress_split.csv · day_type=rain day, cancel_rate | x 100 | ✓ |
| F-33 | 37,530 → 11,063 | outputs/hub_day_metrics.csv · sum(cancelled) by policy | - | ✓ |
| F-34 | −70.5% | outputs/hub_day_metrics.csv · F-33 | (11,063 / 37,530 - 1) x 100 = -70.52% | ✓ |
| F-35 | 125,715 orders · 9.71 → 9.36 · −3.7% | outputs/stress_split.csv · day_type=ordinary weekday | cost_treat / cost_base - 1 | ✓ |
| F-36 | 184,449 orders · 9.65 → 8.30 · −14.0% | outputs/stress_split.csv · day_type=weekend or Ramadan | cost_treat / cost_base - 1 | ✓ |
| F-37 | 16,817 orders · 10.42 → 6.84 · −34.4% | outputs/stress_split.csv · day_type=rain day | cost_treat / cost_base - 1 | ✓ |
| F-38 | RAMADAN band from Feb 18 | data/days.csv · first date with ramadan = True | - | ✓ |

Assumptions: A-02, A-03, A-08, A-10

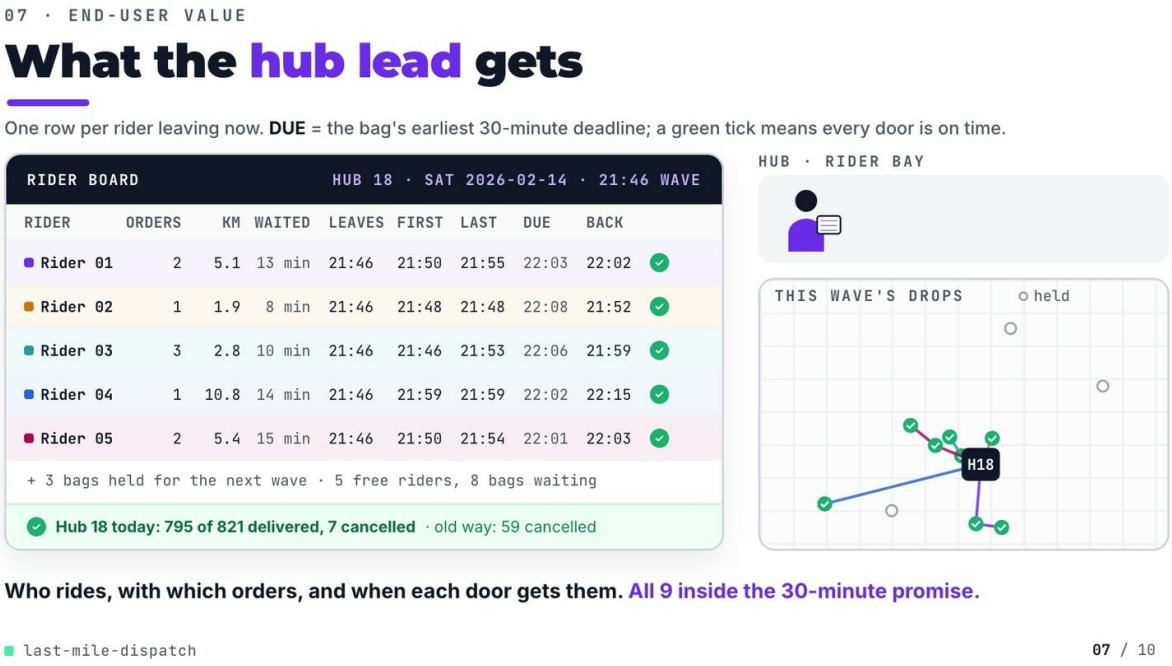
**Illustrative, not data:**

- Chart and bars are drawn from the injected numbers (no hand-typed data points); bars are to scale from zero.

To change this slide: Re-run tools/export_series.py and inject it (--name SERIES); layout in compositions/s04-routes.html.

## Slide 07: End-user value (white)

52.9-65.9 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py` from `tools/export_rider_board.py`)




### On-screen text (verbatim)

07 · END-USER VALUE

What the hub lead gets

One row per rider leaving now. DUE = the bag's earliest 30-minute deadline; a green tick means every door is on time.

RIDER BOARD · HUB 18 · SAT 2026-02-14 · 21:46 WAVE — columns RIDER, ORDERS, KM, WAITED, LEAVES, FIRST, LAST, DUE, BACK

Rider 01 2 5.1 13 min 21:46 21:50 21:55 22:03 22:02 ✓

Rider 02 1 1.9 8 min 21:46 21:48 21:48 22:08 21:52 ✓

Rider 03 3 2.8 10 min 21:46 21:46 21:53 22:06 21:59 ✓

Rider 04 1 10.8 14 min 21:46 21:59 21:59 22:02 22:15 ✓

Rider 05 2 5.4 15 min 21:46 21:50 21:54 22:01 22:03 ✓

+ 3 bags held for the next wave · 5 free riders, 8 bags waiting

✓ Hub 18 today: 795 of 821 delivered, 7 cancelled · old way: 59 cancelled

HUB · RIDER BAY · THIS WAVE'S DROPS · held

Who rides, with which orders, and when each door gets them. All 9 inside the 30-minute promise.

### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-03 | 26 hubs | `outputs/summary.json; data/hubs.csv` · hubs; row count | - | ✓ |
| F-07 | 30 minutes to each door / 30-minute promise | `src/last_mile_dispatch/config.py:27` · SLA_MIN = 30 | - | ✓ |
| F-39 | Rider 01-05 rows | `demo assets/rider_board.json (re-run of the repo's simulator), injected into compositions/value.html` · rows | km = sum of road legs; waited = wave time - oldest order's creation; due = earliest creation + 30 min | ✓ |
| F-40 | 795 of 821 delivered, 7 cancelled · old way: 59 cancelled (old way = FIFO, one order per trip) | `outputs/hub_day_metrics.csv` · hub_id=18, date=2026-02-14, policy in {bagging+matching+geofence, baseline} | - | ✓ |
| F-41 | + 3 bags held for the next wave · 5 free riders, 8 bags waiting | `demo assets/rider_board.json` · bags_held, free_riders, bags_waiting_before | held = bags waiting - riders matched | ✓ |
| F-42 | All 9 inside the 30-minute promise. | `demo assets/rider_board.json` · orders_dispatched; stops[].on_time | every stop ETA - creation <= 30 min | ✓ |
| F-43 | HUB 18 | `outputs/summary.json` · arms.treatment | - | ✓ |

Assumptions: A-03, A-04, A-05, A-08, A-11

**Illustrative, not data:**

- Rider 01-05 are placeholders (no real people); all five are on bikes.
- The hub-lead figure, rider bay, scooter drawings and driving speed are illustrative; drop positions and stop order are real, scaled to fit the panel (uniform scale).
- Grey circles are the drops of the 3 bags held for the next wave.

To change this slide: Re-run tools/export_rider_board.py (hub, date, wave arguments), then tools/inject_sheet.py --name SHEET; layout in compositions/value.html.

## Slide 08: Results

65.6-74.4 s · composition `compositions/s05-results.html (stats array)`




### On-screen text (verbatim)

08 · RESULTS · HUB-LEVEL A/B, 4 WEEKS

13 treated hubs vs 13 control hubs: cheaper and faster.

Treated hubs vs control hubs over 4 weeks. Cost, cancellations and fake marks: lower is better; on time: higher.

COST −10.6% fleet cost per delivered order · AED 9.79 → 8.75

PROMISE +28.3 points more orders inside 30 min · 58.0% → 86.2%

CANCELLATIONS 3.7% cancellation rate, treated hubs · control 11.6%

DOOR CHECK −72.4% fake "not home" marks accepted · 11.7 → 3.2 per 1,000

326,981 orders · on all four, the 95% confidence range of the difference stays clear of zero (Welch t-test)

### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-13 | 4 WEEKS / over 4 weeks / 4 weeks | `outputs/summary.json` · experiment_days = 28 | 28 / 7 = 4 | ✓ |
| F-27 | 13 vs 13 hubs | `outputs/summary.json` · arms.treatment, arms.control | len of each list | ✓ |
| F-44 | −10.6% | `outputs/ab_readout.csv` · metric=cost_per_delivered, rel_diff | x 100 | ✓ |
| F-45 | AED 9.79 → 8.75 | `outputs/ab_readout.csv` · metric=cost_per_delivered, control, treatment | - | ✓ |
| F-46 | +28.3 | `outputs/ab_readout.csv` · metric=on_time_share_of_orders, diff | x 100 | ✓ |
| F-47 | 58.0% → 86.2% | `outputs/ab_readout.csv` · metric=on_time_share_of_orders, control, treatment | x 100 | ✓ |
| F-48 | 3.7% · control 11.6% | `outputs/ab_readout.csv` · metric=cancel_rate, treatment, control | x 100 | ✓ |
| F-49 | −72.4% | `outputs/ab_readout.csv` · metric=fake_accepted_per_1k, rel_diff | x 100 | ✓ |
| F-50 | 11.7 → 3.2 per 1,000 | `outputs/ab_readout.csv` · metric=fake_accepted_per_1k, control, treatment | - | ✓ |
| F-51 | 326,981 orders | `outputs/summary.json` · orders_in_experiment | - | ✓ |
| F-52 | on all four, the 95% confidence range of the difference stays clear of zero (Welch t-test) | `outputs/ab_readout.csv` · welch_lo, welch_hi | lo > 0 or hi < 0 for cost, on-time, cancel, fake | ✓ |

Assumptions: A-02, A-03, A-04, A-06, A-09, A-12

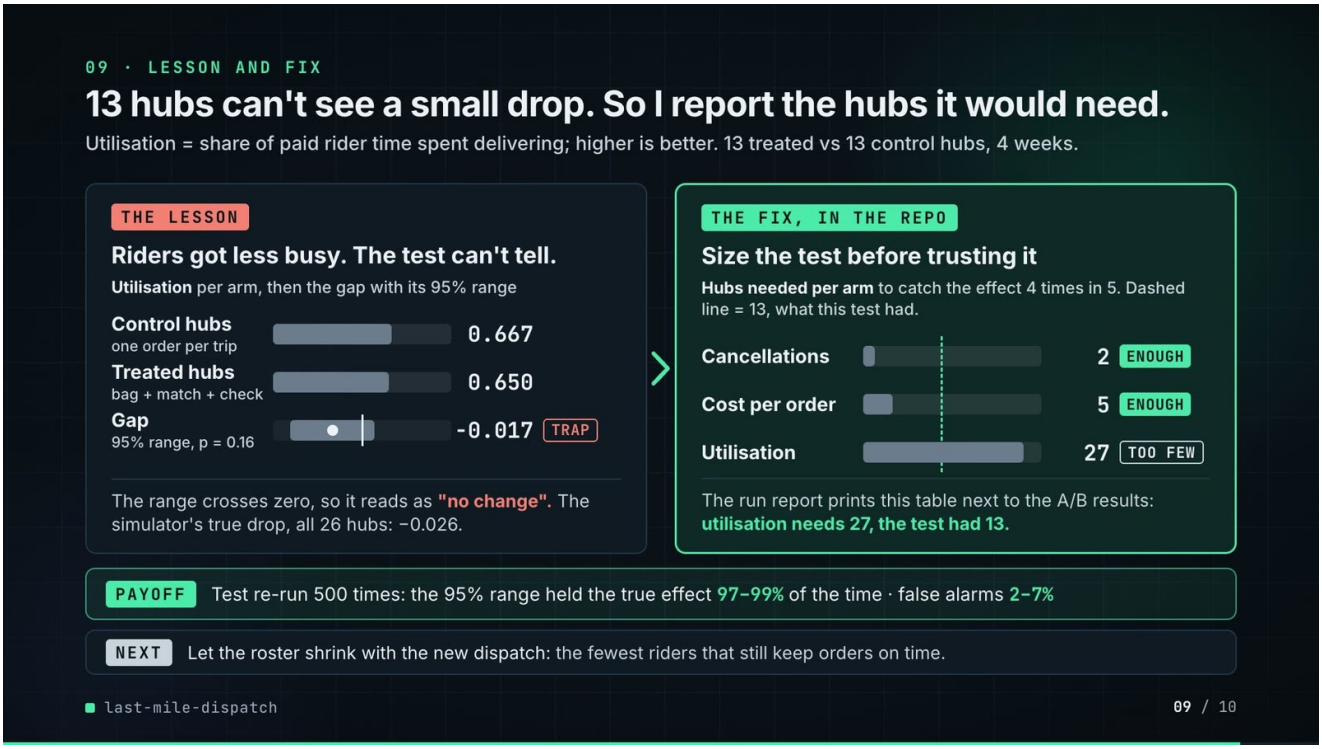
**Illustrative, not data:**

- Mini bars under each number are drawn to the same ratio (treatment / control).
- All four cards are mint because all four moved the good way; the door-check card was coral before this revision.

To change this slide: The stats array and card text in compositions/s05-results.html.

## Slide 09: Lesson and fix

74.0-83.0 s · composition `compositions/s06-twist.html`




### On-screen text (verbatim)

09 · LESSON AND FIX

13 hubs can't see a small drop. So I report the hubs it would need.

Utilisation = share of paid rider time spent delivering; higher is better. 13 treated vs 13 control hubs, 4 weeks.

THE LESSON · Riders got less busy. The test can't tell. · Utilisation per arm, then the gap with its 95% range

Control hubs (one order per trip) 0.667 · Treated hubs (bag + match + check) 0.650 · Gap (95% range, p = 0.16) −0.017 TRAP

The range crosses zero, so it reads as "no change". The simulator's true drop, all 26 hubs: −0.026.

THE FIX, IN THE REPO · Size the test before trusting it · Hubs needed per arm to catch the effect 4 times in 5. Dashed line = 13, what this test had.

Cancellations 2 ENOUGH · Cost per order 5 ENOUGH · Utilisation 27 TOO FEW

The run report prints this table next to the A/B results: utilisation needs 27, the test had 13.

PAYOFF · Test re-run 500 times: the 95% range held the true effect 97–99% of the time · false alarms 2–7%

NEXT · Let the roster shrink with the new dispatch: the fewest riders that still keep orders on time.

### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-13 | 4 WEEKS / over 4 weeks / 4 weeks | `outputs/summary.json` · experiment_days = 28 | 28 / 7 = 4 | ✓ |
| F-26 | 500 re-draws / Test re-run 500 times | `outputs/rerandomisation.csv` · draws | - | ✓ |
| F-27 | 13 vs 13 hubs | `outputs/summary.json` · arms.treatment, arms.control | len of each list | ✓ |
| F-53 | 0.667 / 0.650 | `outputs/ab_readout.csv` · metric=utilization, control, treatment | - | ✓ |
| F-54 | gap strip from -0.040 to +0.007, crossing zero | `outputs/ab_readout.csv` · metric=utilization, welch_lo, welch_hi | - | ✓ |
| F-55 | 95% range, p = 0.16 | `outputs/ab_readout.csv` · metric=utilization, welch_p | - | ✓ |
| F-56 | 27 · TOO FEW / utilisation needs 27, the test had 13 | `outputs/power.csv` · metric=utilization, hubs_per_arm_needed | - | ✓ |
| F-57 | track -0.05..+0.05 = 260 px; range 25 -> 148 px (left 25, width 123), estimate dot centred at 87 px (left 75), zero at 130 px | `compositions/s06-twist.html`; `outputs/ab_readout.csv` · welch_lo, welch_hi, diff | px = (v + 0.05) x 2,600 | ✓ |
| F-62 | -0.017 | `outputs/ab_readout.csv` · metric=utilization, diff | - | ✓ |
| F-63 | The simulator's true drop, all 26 hubs: -0.026 | `outputs/ab_readout.csv` · metric=utilization, true_effect_all_hubs | - | ✓ |

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-64 | Cancellations 2 · Cost per order 5 · ENOUGH | outputs/power.csv · metric in {cancel_rate, cost_per_delivered}, hubs_per_arm_needed | ENOUGH = needed <= 13 | ✓ |
| F-65 | to catch the effect 4 times in 5 | src/last_mile_dispatch/experiment.py:77, :85 · power: float = 0.8 | - | ✓ |
| F-66 | 97-99% of the time · false alarms 2-7% | outputs/rerandomisation.csv · welch_coverage, aa_false_positive_welch | min and max over metrics, x 100, rounded | ✓ |
| F-67 | THE FIX, IN THE REPO · Size the test before trusting it · The run report prints this table next to the A/B results | src/last_mile_dispatch/experiment.py:77-90, :127; cli.py:78-98; tests/test_experiment.py:47, :53, :66; outputs/run_report.md · functions, tests, report sections | - | ✓ |
| F-68 | Dashed line = 13, what this test had | outputs/summary.json · len(arms.treatment) | - | ✓ |
| F-69 | NEXT · Let the roster shrink with the new dispatch: the fewest riders that still keep orders on time. | README 'Insights and learning' (last paragraph) and 'Known limitations' (the biggest gap: the roster is fixed) · README text | - | ✓ |

Assumptions: A-02, A-09, A-12

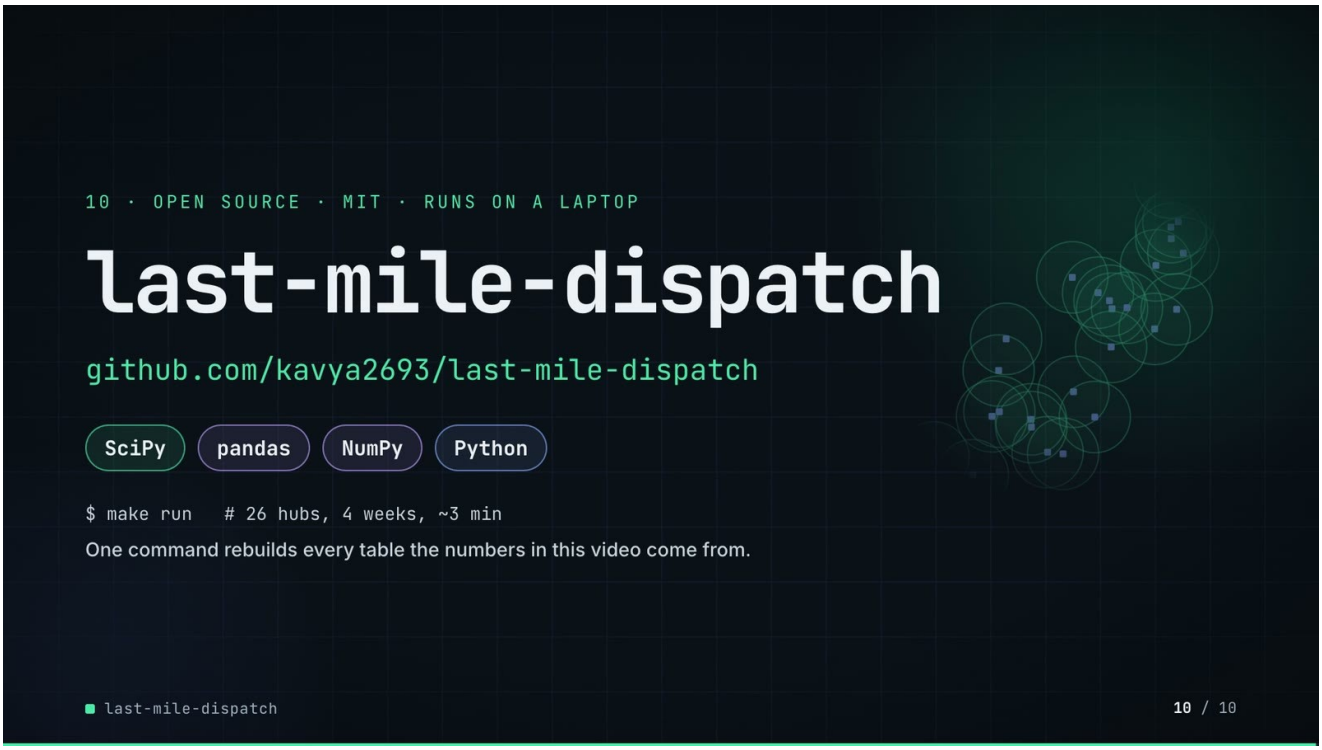
**Illustrative, not data:**

- Left panel: utilisation bars 1.0 = 260 px from zero; the gap strip spans -0.05..+0.05 over the same 260 px with zero marked.
- Right panel: hubs-per-arm bars 30 hubs = 260 px; the two panels show different units, so they do not share one scale.
- Bars are neutral grey; coral marks only the TRAP (a range that crosses zero read as 'no change'), mint marks the fix.

To change this slide: Panels #twist-left / #twist-right, the payoff and next strips in compositions/s06-twist.html; numbers from outputs/ab_readout.csv, power.csv, rerandomisation.csv.

## Slide 10: End card

82.6-87.0 s · composition `compositions/s07-end.html`




### On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP

last-mile-dispatch

github.com/kavya2693/last-mile-dispatch

SciPy · pandas · NumPy · Python

$ make run  # 26 hubs, 4 weeks, ~3 min

One command rebuilds every table the numbers in this video come from.

### Figures on this slide

| ID | Shown | Source · key | Calculation | Check |
|---|---|---|---|---|
| F-11 | hub dots + circles | `data/hubs.csv; config.py:12 (SERVICE_RADIUS_KM = 4.0)` · lat/lon -> km from the centroid (demo assets/run_series.json) | - | ✓ |
| F-58 | MIT | `LICENSE line 1` · - | - | ✓ |
| F-59 | ~3 min | `outputs/summary.json` · runtime_s = 179.8 | 179.8 / 60 = 3.0 min | ✓ |
| F-60 | SciPy · pandas · NumPy · Python | `pyproject.toml dependencies` · numpy, pandas, scipy | - | ✓ |
| F-61 | $ make run  # 26 hubs, 4 weeks, ~3 min | `Makefile; outputs/summary.json` · run target; hubs; experiment_days | - | ✓ |

Assumptions: A-12

**Illustrative, not data:**

- Faded background: the 26 real hub positions with 4 km circles.

To change this slide: Text in compositions/s07-end.html.

## 5. Change-impact matrix

| Input / assumption | Affected figures | Slides | Files to edit | Re-run |
|---|---|---|---|---|
| A-01: All data is synthetic, produced by a seeded generator sized to the production problem (26 hubs, 473 riders, 42 days, 477,868 orders) | all figures | 01, 02, 03, 04, 05, 06, 07, 08, 09, 10 | `scripts/generate_data.py`; `src/last_mile_dispatch/config.py` | make data; make run |
| A-02: The rider roster is fixed and sized for an ordinary weekday evening; every policy pays for the same riders | F-04, F-33, F-34, F-35, F-36, F-37, F-44, F-45, F-53, F-54, F-55, F-56, F-62, F-63, F-69 | 02, 05, 06, 08, 09 | `scripts/generate_data.py:84-85` | make data; make run |
| A-03: Travel is straight-line distance x 1 | F-32, F-33, F-34, F-35, F-36, F-37, F-39, F-44, F-45, F-46, F-47 | 06, 07, 08 | `src/last_mile_dispatch/config.py:13, :36`; `src/last_mile_dispatch/geo.py:29-41` | make run; export_rider_board.py + inject_sheet.py |
| A-04: Matching prices each rider-bag pair in AED: km cost of the vehicle, 1 | F-14, F-33, F-34, F-39, F-44, F-45, F-46, F-47, F-48 | 03, 06, 07, 08 | `src/last_mile_dispatch/config.py:74-76` | make run; export_rider_board.py + inject_sheet.py |
| A-05: Bagging limits: drops within 1 km and created within 12 minutes of the seed order, at most 3 stops, and the first customer may wait at most 9 extra minutes | F-15, F-16, F-17, F-39, F-41 | 04, 05, 07 | `src/last_mile_dispatch/config.py:37, :65-67` | make run; export_rider_board.py + inject_sheet.py |
| A-06: Geofence rule: a 'customer unavailable' mark is flagged when fewer than 2 of the rider's last 6 GPS pings are within 150 m of the customer; flagged riders are sent back to the door (treatment) or the flag is only logged (baseline) | F-21, F-25, F-49, F-50 | 05, 08 | `src/last_mile_dispatch/config.py:79-80` | make run |
| A-07: Fake marks: about 8% of riders (36 of 473 in the data) mark 'customer unavailable' without reaching the door on 22% of their stops; fakes are known by construction and riders do not change habit when flagged | F-10, F-21, F-49, F-50 | 02, 05, 08 | `src/last_mile_dispatch/config.py:40-41` | make data; make run |
| A-08: Service promise 30 minutes from order creation; a customer cancels if nothing has left the hub after 45 minutes; the dispatcher wakes every 2 minutes; 2 minutes handover per stop | F-07, F-08, F-24, F-32, F-33, F-34, F-42, F-46, F-47, F-48 | 02, 03, 05, 06, 07, 08 | `src/last_mile_dispatch/config.py:23-29` | make run; export_rider_board.py + inject_sheet.py |
| A-09: Experiment design: hubs sorted by pre-period orders per day, paired, one coin flip per pair (seed 2026); one number per hub over four weeks; Welch t and hub-bootstrap 95% intervals; power from the pre-period between-hub spread | F-27, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-52, F-53, F-54, F-55, F-56, F-57, F-62, F-63, F-64, F-65, F-66, F-67, F-68 | 05, 08, 09 | `src/last_mile_dispatch/cli.py:22`; `src/last_mile_dispatch/experiment.py` | make run |
| A-10: Calendar: two pre-period weeks from 2026-01-19, then four experiment weeks 2026-02-02 to 2026-03-01; weekends are Saturday-Sunday; Ramadan from 2026-02-18; one rain day 2026-02-07 (35% more orders, 40% slower traffic) | F-06, F-09, F-13, F-31, F-32, F-35, F-36, F-37, F-38, F-40 | 02, 03, 05, 06, 07, 08, 09 | `src/last_mile_dispatch/config.py:16-18`; `scripts/generate_data.py:35-36` | make data; make run |
| A-11: The rider board (slide 07) re-runs the repo's own simulator for hub 18 on 2026-02-14 under the treatment and records the 21:46 wave; drop ETAs replay the simulator's clock with geo | F-39, F-40, F-41, F-42 | 07 | `docs/demo/source/tools/export_rider_board.py (hub, date, wave arguments)` | export_rider_board.py + inject_sheet.py; hyperframes render |
| A-12: Video rounding: percentages to 1 decimal, AED to 2 decimals, utilisation and CI bounds to 3 decimals, counts with thousands separators; '~3 min' and '180 s' are rounded run times | F-16, F-17, F-23, F-32, F-34, F-35, F-36, F-37, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-53, F-54, F-55, F-59, F-62, F-63, F-66 | 04, 05, 06, 08, 09, 10 | `the composition file named on each slide` | hyperframes render |
| A-13: Slide 01 (industries, everyday problems) is general delivery-operations knowledge used as context | F-01, F-02 | 01 | `compositions/s01-context.html` | hyperframes render |

## 6. Known differences and caveats

• Everything is simulated (A-01). The A/B readout is what an experimenter would see; the 'true effect' is known only because every policy ran on every hub-day with the same random draws.

• The cost saving is mostly cancellations turned into deliveries on a fixed, paid roster (A-02); slide 06 shows it is under 4% on an ordinary weekday and 34% on the rain day.

• Slide 07's ETAs are reconstructed with the repo's travel function for one wave; they equal the simulator's delivery times because the five riders are honest and all nine customers are home (asserted in export_rider_board.py).

• Slide 08 uses hub-level means (13 vs 13); the ablation numbers on slide 04 pool every hub and every day, so they differ slightly from the A/B means.

• '36 riders sometimes skip doors' is a generator setting (8% of riders, 22% of their stops), not an observed behaviour (A-07).

• Slide 01 statements are general industry knowledge (A-13).

• Rider names on slide 07 are placeholders.

• Slide 09 shows the one disappointing result the repo can do something about in code (an underpowered metric), and the fix it implements (pre-period power, hubs-per-arm table, re-randomisation check). The fix does not raise utilisation itself; that needs a smaller roster, shown as NEXT and listed in README 'Known limitations'.

• Other non-wins the slide drops are still reported in the README 'What didn't win' row and 'The result': the 3.7% weekday saving (also on slide 06), matching alone adding about half what bagging does, and the geofence missing 31.9% of kerb marks while flagging 5.3% of honest riders in towers.

• The payoff range (97-99% coverage, 2-7% false alarms) spans the seven A/B metrics in outputs/rerandomisation.csv; utilisation itself had 97.6% coverage and only 62% power at 13 hubs per arm.

## Appendix: environment

| Item | Value |
|---|---|
| Commit | `c7de8a6ad78ef3961f773e55ae9af197085c3f9c` |
| Python | 3.11.7 |
| numpy | 2.4.6 |
| pandas | 3.0.6 |
| scipy | 1.17.1 |
| Profile / seeds | data=7, hub_assignment=2026, rerandomisation=2027, simulator=hub_id * 10000 + day index (same for every policy) |
| Run time | summary.json runtime_s = 179.8 s (about 3 min) for make run with 4 workers on the author's laptop |